

Smell & Taste

5000 olfactory - Volatile odorants

Cilia receptors which recognize odorants and amplify them

Odorant receptors → G-protein coupled receptor

other receptors → T2Rs (detect amines)
eg. amino acids

each odorant is combined by a unique combination of receptors -

each olfactory sensory neuron expresses only one odorant receptor gene and one type of receptor

each odorant is detected by different types of receptors

each odorant is detected by a unique combination of receptors

• Neurons with ≠ receptors are interspersed

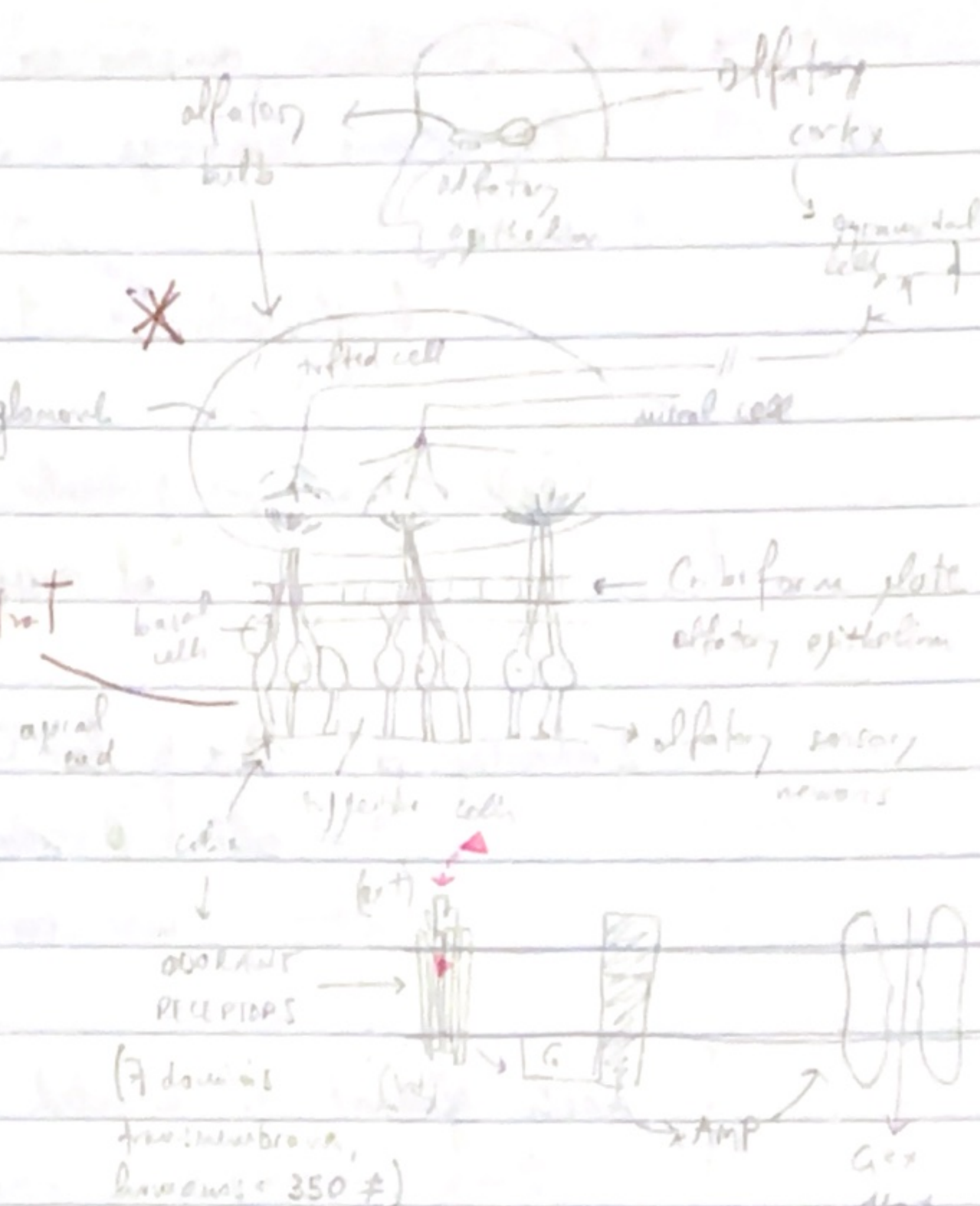
⇒ The axons of olfactory sensory neurons project to the ipsilateral olfactory bulb.

They make synapses with 3 types of neurons: - mitral and tufted relay neurons
- periglomerular interneurons

• Axon of 1 neuron → 1 glomerulus.

• The young dendrite of mitral and tufted → 1 glomerulus.

1 glomerulus → + 1000 sensory neurons
40 - 50 relay neurons.



⊕ odorant receptor → G-protein → cAMP → cation influx
↓
depolarization

→ In the epithelium, receptors are scattered everywhere. In the bulb, their axons converge in a few glomeruli at two spots (each side)

1 glomerulus ^{receptor input} → 1 type of receptor

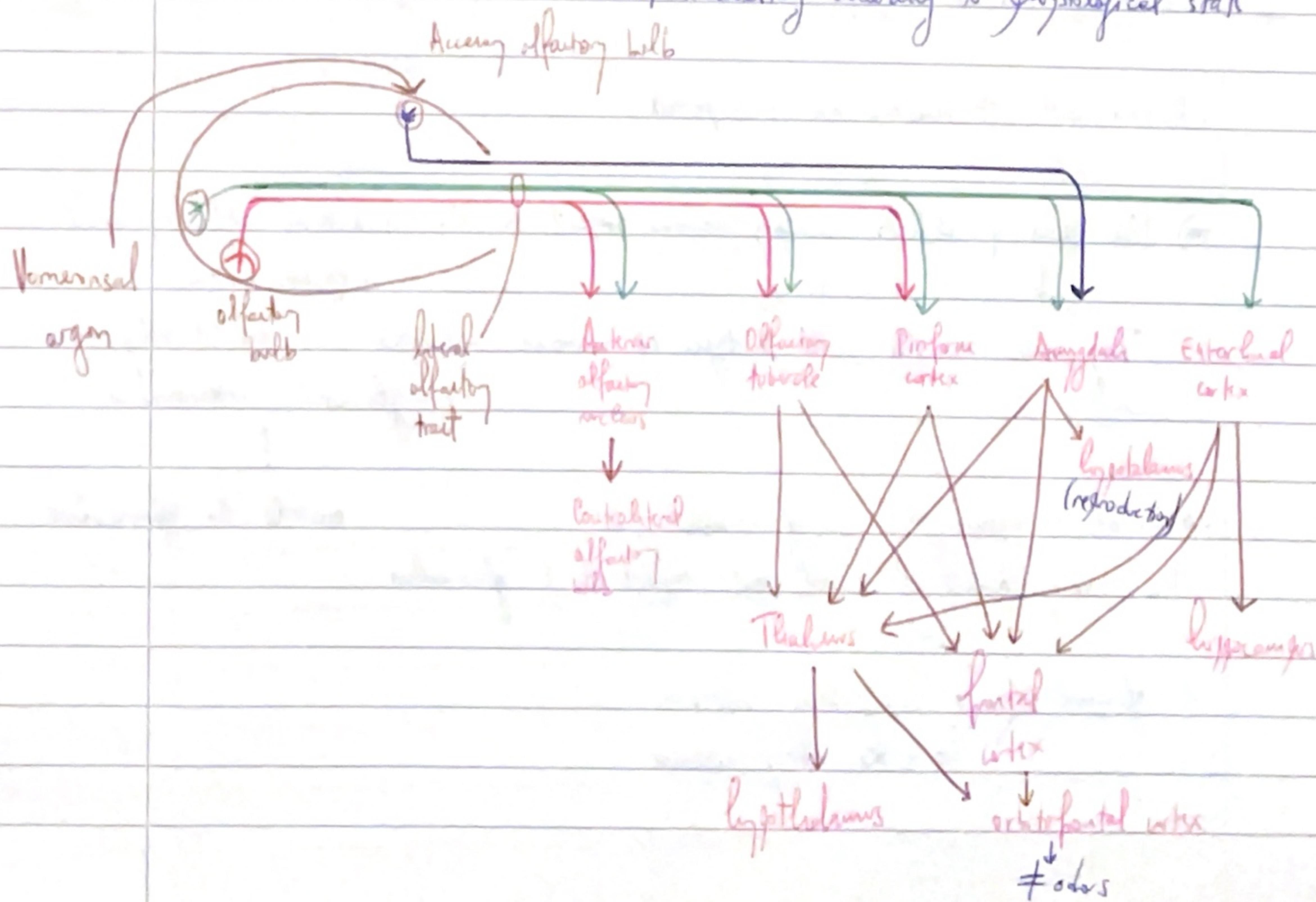
each odorant → particular combination of receptor types → particular combination of glomeruli

2 advantages → Detects low [] of odorant
cells change in a short grad of time but the code remains in the same place

each glomerulus is enveloped by periglomerular interneurons.
modulate → inhibit some dendrites.

Cranial cell receive input of basal dendrites → inhibit some postsynaptic neurons
filter stimulus.

→ 2 retrograde projections from the olfactory cortex (basal forebrain & midbrain) help modulating according to physiological state



The olfactory system also detect pheromones → substances that influence the behaviour of others of the same species.

(nasal olfactory epithelium)
(vomeronasal organ)
(lost in humans)
chemosensory receptors (V1R + V2R)

Taste

5 submodalities: sweet, bitter, salty, sour, umami
carbohydrate (amino acids) protein
gustatory glands → "Tastants"

• taste → these 5

• flavor → mix of taste, olfactory and somatosensory inputs

⇒ Tastants are detected by taste receptor cells that are clustered in taste buds, gaguli gustation

⇒ Most taste buds are on the tongue but also in other places.

⇒ In the tongue, they occur in papillae
fingiform → 1-5 taste bud
foliate } hundreds of taste buds
circumvallate }

⇒ In the epithelium, the taste pore allows the contact with tastants through microvilli

⇒ 1 taste bud ~ 100 taste receptor cells.

↑
elongated cells → supportive
round cells → stem cells

Taste cell
↑ non-neural but "can do" chemical synapses and electrically excitable
↑ basal end → afferent fibers of gustatory sensory neurons

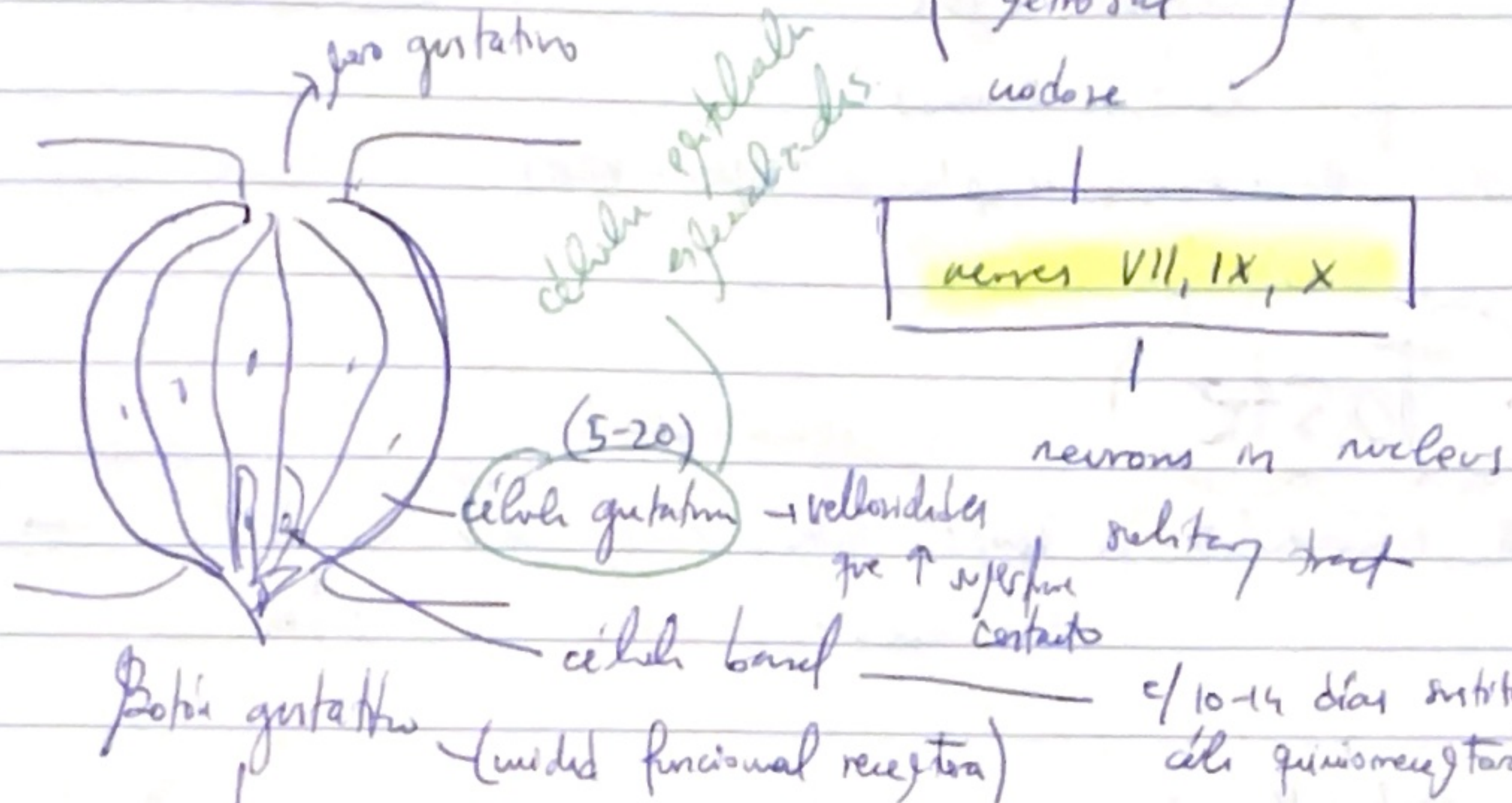
electrically excitable → voltage gated Na^+ , K^+ , Ca^{2+} channels.

⇒ Each taste is detected by a distinct sensory Transduction Mechanism and distinct population of taste cells.

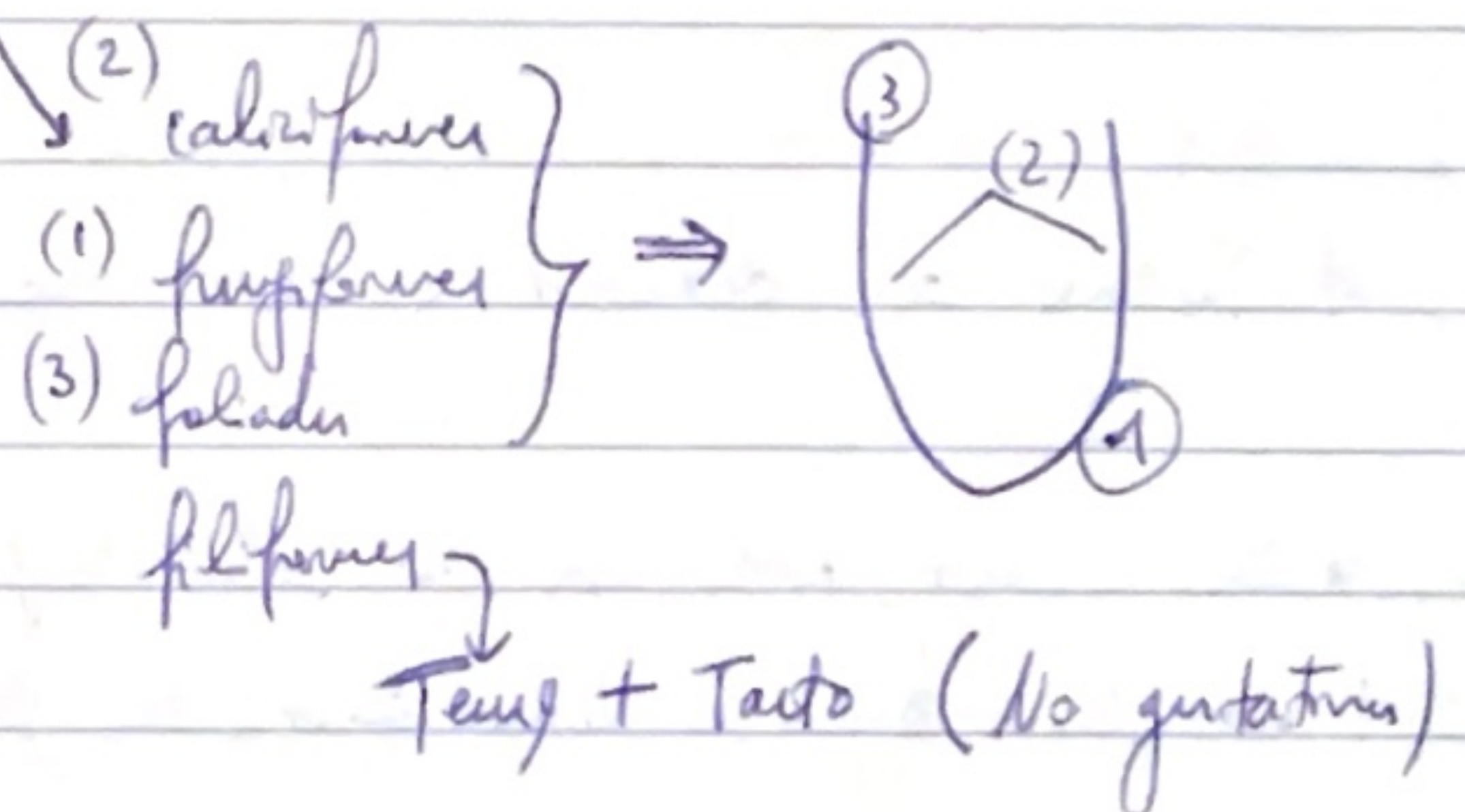
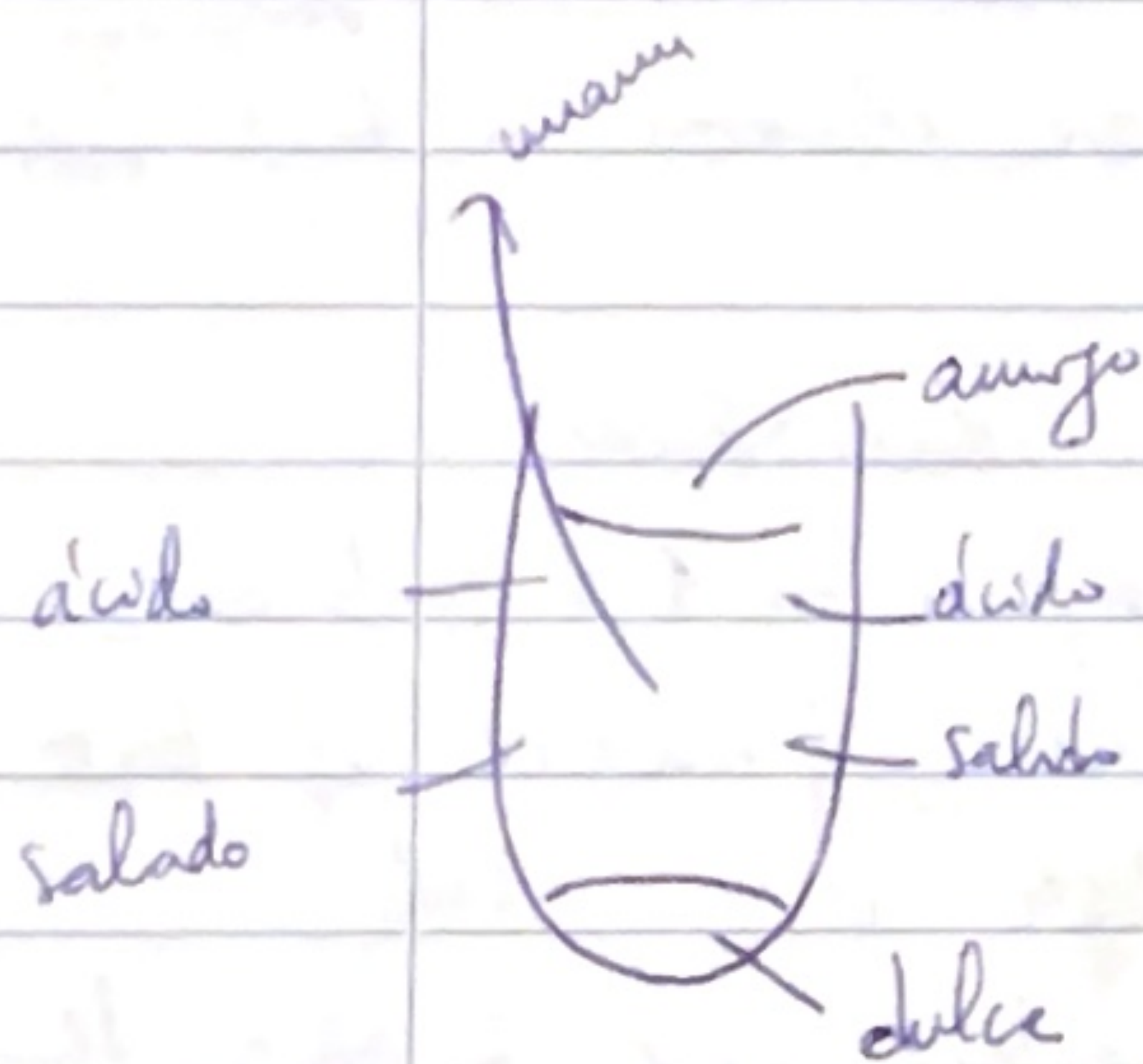
↑ bitter, sweet, umami → G-protein-coupled receptors
salty, sour → Ion channels.

Taste cells → neurotransmitter → Sensory fibers → Action potential
 signal to sensory cell body

cell bodies of gustatory sensory neurons → geniculate } ganglia
 retro-sal }
 nodose



en la papila gustativa — dorso + borde lateral lengua
 paladar blando
 faringe



VII → parte anterior lengua
 IX → " posterior y paladar
 X → parte faringe

Núcleo del tracto solitario

